6793 - Spectroscopic Verification of Robust z>15 Galaxy Candidates Selected with Multiple Medium-Band Datasets

Cycle: 4, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Yuichi Harikane (PI)	**University of Tokyo, Institute of Cosmic Ray Research**
Prof. Masami Ouchi (CoI)	National Astronomical Observatory of Japan (NAOJ)
Dr. Yoshiaki Ono (CoI)	University of Tokyo, Institute of Cosmic Ray Research
Dr. Kimihiko Nakajima (CoI)	Kanazawa University
Dr. Makoto Ando (CoI)	University of Tokyo, Institute of Cosmic Ray Research

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	gds_v1	NIRSpec MultiObject Spectroscopy	(1) catalog_gds
	2	gdn_v7_test1	NIRSpec MultiObject Spectroscopy	(8) catalog_gdn_all_v4
	3	a2744all_v1	NIRSpec MultiObject Spectroscopy	(3) catalog_a2744all
	4	gdn_v8_all_v3	NIRSpec MultiObject Spectroscopy	(10) catalog_gdn_all_v6

ABSTRACT

JWST was designed to study the first galaxies, which are theoretically thought to have emerged at z~20-30. Since its first operation, the JWST has discovered a surprisingly large number of galaxies at z > 10, and its NIRSpec spectroscopy has successfully confirmed some of these galaxies up to z~14. Because z~14 is the upper limit for galaxies selected as F150W-dropouts, the next logical step is to target even higher redshift candidates, F200W-dropout galaxies at z~15-20. However, efforts to spectroscopically confirm F200W-dropouts have faced challenges. For example, NIRSpec observations showed that a previously claimed z~16 galaxy candidate was actually a foreground object at z=4.9 whose strong emission lines boosted

fluxes in broad and medium-bands, making a SED similar to that of a z~16 galaxy.

In this program, we will conduct NIRSpec/PRISM spectroscopy targeting 10 carefully selected F200W-dropout galaxies at z~15-20. These galaxies are chosen based on extensive datasets that include multiple medium-band data crucial for removing interlopers with strong lines identified in the previous F200W-dropout selection, and multi-epoch broad-band data useful to exclude transient contamination, ensuring that our targets are robust z>15 candidates. Successful confirmation of their redshifts would not only set a new redshift record but also provide a unique opportunity to study the star formation activity and the physical properties of galaxies at z>15. Alternatively, if all of our candidates are revealed to be interlopers, this would indicate a sharp decline of the cosmic SFR density from z~12 to z>15, suggesting a rapid buildup of the first galaxies ~200 Myrs after the Big Bang.

OBSERVING DESCRIPTION

We propose NIRSpec spectroscopy targeting 10 F200W-dropout galaxies at z~15-20, selected with multiple medium-band data and multi-epoch broadband data. We will use NIRSpec/PRISM to detect Lyman-alpha breaks of these candidates. We will use MOS to efficiently observe multiple targets in one pointing. Our exposure strategy follows previous successful observations.

Proposal 6793 - Targets - Spectroscopic Verification of Robust z>15 Galaxy Candidates Selected with Multiple Medium-Band Datasets

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	catalog_gds	RA: 03 32 30.0120 (53.1250500d) Dec: -27 47 27.71 (-27.79103d) Equinox: J2000		
Comments: *Description=[]*				
(3)	catalog_a2744all	RA: 00 14 8.9640 (3.5373500d) Dec: -30 22 17.44 (-30.37151d) Equinox: J2000		
Comments: *Description=[]*				
(6)	catalog_gdn_all_v2_onlyhighP	RA: 12 36 45.7440 (189.1906000d) Dec: +62 14 11.08 (62.23641d) Equinox: J2000		
Comments: *Description=[]*				
(8)	catalog_gdn_all_v4	RA: 12 36 47.1288 (189.1963700d) Dec: +62 15 12.60 (62.25350d) Equinox: J2000		
Comments: *Description=[]*				
(9)	catalog_gdn_all_v5	RA: 12 36 47.0299 (189.1959579d) Dec: +62 15 15.12 (62.25420d) Equinox: J2000		
Comments: *Description=[]*				
(10)	catalog_gdn_all_v6	RA: 12 36 47.0299 (189.1959579d) Dec: +62 15 15.12 (62.25420d) Equinox: J2000		
Comments: *Description=[]*				
(11)	catalog_gdn_all_v7	RA: 12 36 47.2056 (189.1966900d) Dec: +62 15 14.76 (62.25410d) Equinox: J2000		
Comments: *Description=[]*				

Observation

Proposal 6793, Observation 1: gds_v1

Fri Feb 06 20:00:34 GMT 2026

Diagnostic Status: Error

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(gds_v1 (Obs 1)) Error (Form): This observation was created with an Aperture PA of 30.0000 but it has been assigned an Aperture PA of 86.3643

(Aperture PA) Error (Form): This observation was created with an Aperture PA of 30.0000 but it has been assigned an Aperture PA of 86.3643

(Visit 1:1) Error (Form): Reference stars are required but none were found for this visit

(gds_v1 (Obs 1)) Warning (Form): Config c1 (#1) has 1 master background shutters affected by failed open or closed shutters.

(gds_v1 (Obs 1)) Warning (Form): Config c1 (#2) has 1 master background shutters affected by failed open or closed shutters.

(gds_v1 (Obs 1)) Warning (Form): Config c1 (#3) has 1 master background shutters affected by failed open or closed shutters.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	catalog_gds	RA: 03 32 30.0120 (53.1250500d) Dec: -27 47 27.71 (-27.79103d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	catalog_gds (5 sources)		jwst-nirspec-prism	1.5

Reference Stars

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	53.123814333333 33 Degrees -27.802262222222 225 Degrees	86.364886608800 02			3	18	26522.602
2	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	53.123814333333 33 Degrees -27.802262222222 225 Degrees	86.364886608800 02			3	18	26522.602
3	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	53.123814333333 33 Degrees -27.802262222222 225 Degrees	86.364886608800 02			3	18	26522.602

Special Requirements	
	MSA Scheduled Aperture PA 86.3643 to 86.3643 Degrees (V3 307.78976 to 307.78976)

Observation

Proposal 6793, Observation 2: gdn_v7_test1

Fri Feb 06 20:00:34 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(gdn_v7_test1 (Obs 2)) Warning (Form): Config c1 (#1) has 1 primary slit traces affected by failed open shutters.
(gdn_v7_test1 (Obs 2)) Warning (Form): Config c1 (#2) has 1 primary slit traces affected by failed open shutters.
(gdn_v7_test1 (Obs 2)) Warning (Form): Config c1 (#3) has 1 primary slit traces affected by failed open shutters.
(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(8)	catalog_gdn_all_v4	RA: 12 36 47.1288 (189.1963700d) Dec: +62 15 12.60 (62.25350d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	Filter: CLEAR; Readout: NRSRAPIDD6; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	sci (7340 sources)		jwst-nirspec-prism	1.5

Reference Stars

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
1	935431	189.178410	62.238060	24.84	1	947119	189.182770	62.264930	24.87
1	937958	189.166430	62.242870	24.99	1	948445	189.112210	62.269870	24.65
1	939073	189.188530	62.244930	24.87	1	948778	189.109240	62.271070	24.13
1	942528	189.179970	62.251860	24.94	1	960337	189.140360	62.301820	23.48

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	189.17270566666664 Degrees 62.27571972222222 Degrees	38.859105367204926			3	18	26522.602
2	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	189.17270566666664 Degrees 62.27571972222222 Degrees	38.859105367204926			3	18	26522.602
3	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	189.17270566666664 Degrees 62.27571972222222 Degrees	38.859105367204926			3	18	26522.602

Special Requirements	
	Aperture PA Range 38.8800 to 38.8800 Degrees (V3 260.3054303 to 260.3054303) [MSA Selected] MSA Scheduled Aperture PA 38.8800 to 38.8800 Degrees (V3 260.3054303 to 260.3054303)

Observation

Proposal 6793, Observation 3: a2744all_v1

Fri Feb 06 20:00:34 GMT 2026

Diagnostic Status: Error

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(a2744all_v1 (Obs 3)) Error (Form): This observation was created with an Aperture PA of 30.0000 but it has been assigned an Aperture PA of 180.3786

(Aperture PA) Error (Form): This observation was created with an Aperture PA of 30.0000 but it has been assigned an Aperture PA of 180.3786

(Visit 3:1) Error (Form): Reference stars are required but none were found for this visit

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	catalog_a2744all	RA: 00 14 8.9640 (3.5373500d) Dec: -30 22 17.44 (-30.37151d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	catalog_a2744all (2 sources)		jwst-nirspec-prism	1.5

Reference Stars

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	3.536227625 Degrees -30.382716388888866 Degrees	180.3792167588352			3	18	26522.602
2	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	3.536227625 Degrees -30.382716388888866 Degrees	180.3792167588352			3	18	26522.602
3	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	3.536227625 Degrees -30.382716388888866 Degrees	180.3792167588352			3	18	26522.602

Special Requirements	
	MSA Scheduled Aperture PA 180.3786 to 180.3786 Degrees (V3 41.804066 to 41.804066)

Observation

Proposal 6793, Observation 4: gdn_v8_all_v3

Fri Feb 06 20:00:34 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(10)	catalog_gdn_all_v6	RA: 12 36 47.0299 (189.1959579d) Dec: +62 15 15.12 (62.25420d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	Filter: CLEAR; Readout: NRSRAPIDD6; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	sci (7328 sources)		jwst-nirspec-prism	1.5

Reference Stars

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
1	946089	189.096070	62.261610	24.59	1	956095	189.190600	62.290670	25.42
1	948445	189.112210	62.269870	24.65	1	958474	189.084200	62.297070	24.71
1	949535	189.123010	62.273410	25.47	1	959490	189.116050	62.299740	24.09
1	952472	189.169860	62.281910	24.28	1	959816	189.105480	62.300300	25.25

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	189.133281125 Degrees 62.295753055555 55 Degrees	340.70478712227 81			3	18	26522.602
2	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	189.133281125 Degrees 62.295753055555 55 Degrees	340.70478712227 81			3	18	26522.602
3	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	189.133281125 Degrees 62.295753055555 55 Degrees	340.70478712227 81			3	18	26522.602

Special Requirements	
	MSA Scheduled Aperture PA 340.7602 to 340.7602 Degrees (V3 202.18562 to 202.18562)